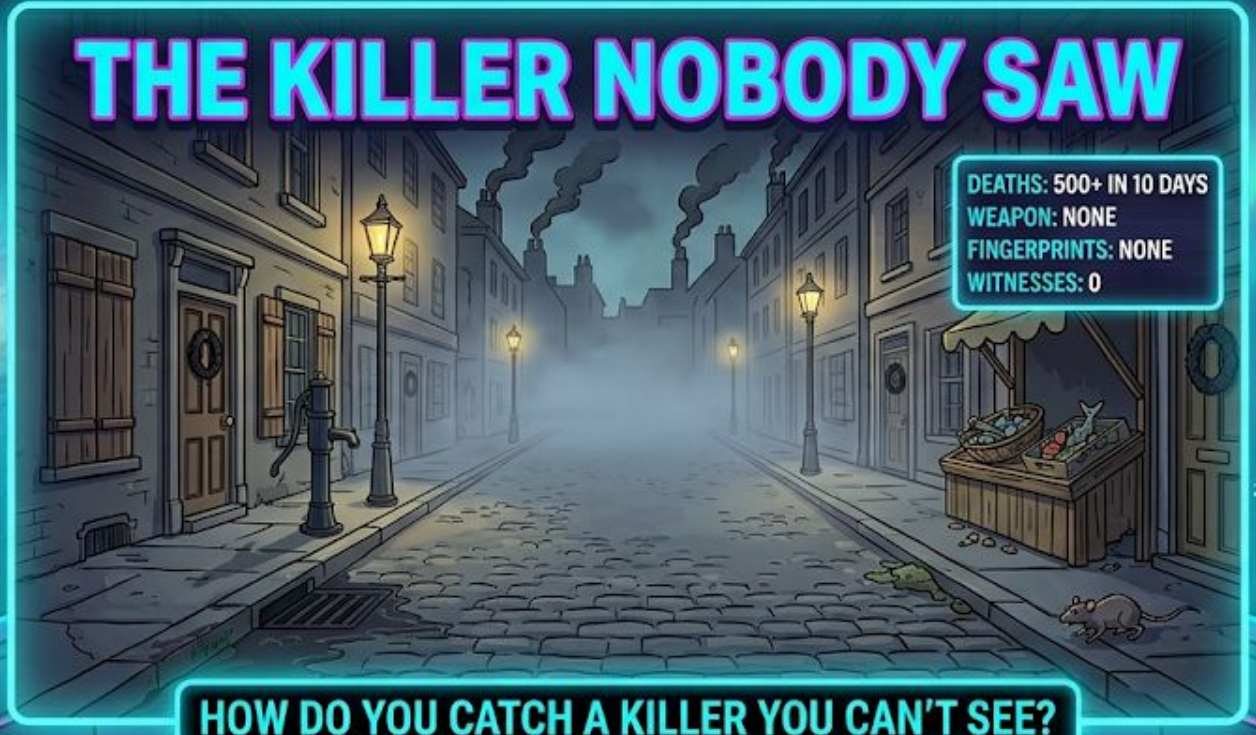


// SELECT SIMULATION → EPIDEMIC FORENSICS


EPIDEMIC FORENSICS
CASE #1 · SOHO, LONDON · 1854
AVATAR: DR. INFECTIONISTA

THE KILLER NOBODY SAW

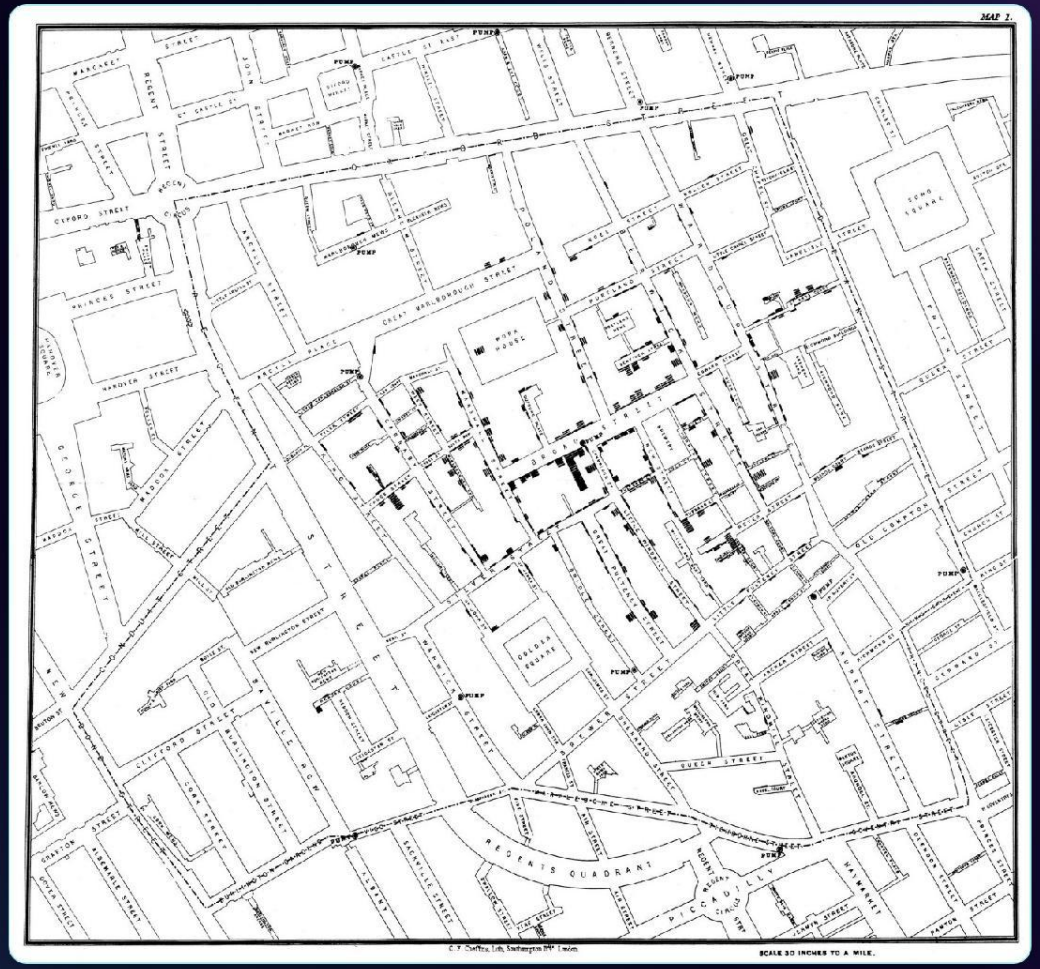


HOW DO YOU CATCH A KILLER YOU CAN'T SEE?

CANVAS · STAGE 01

Theories: who is the killer — and why can't anyone catch it?

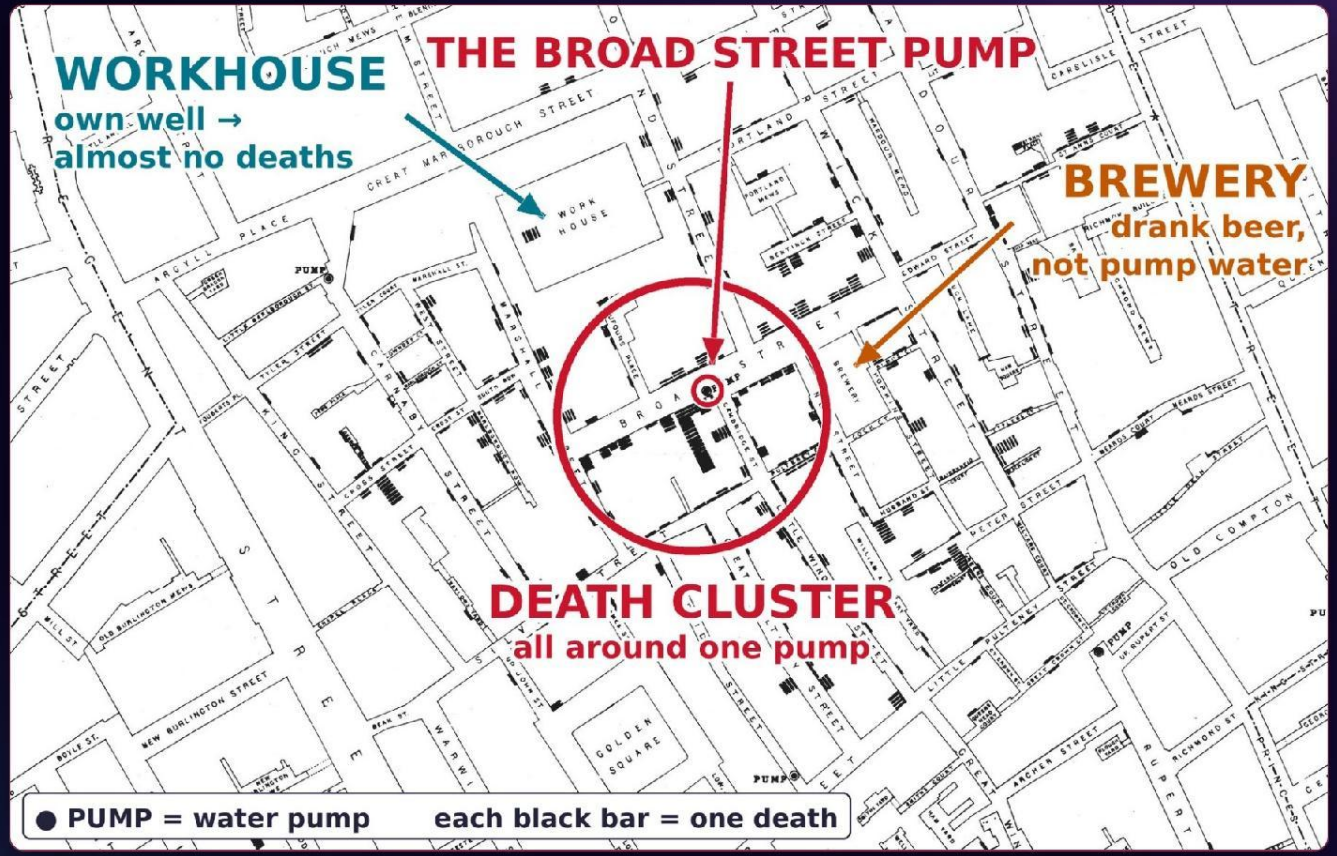
· for drawings and notes ·



Each small **black bar** = one death. Black dots marked **PUMP** = water pumps.

Detective: where are the deaths thickest? Where did the killer leave a trail?

▸ answer on the next slide



The deaths converge on **one pump in Broad Street**. The workhouse and the brewery sat right next to it — but had their own water, so almost nobody there died. Later it turned out the well had been dug about three feet from a leaking cesspit.

CANVAS · STAGE 02

Circle the cluster. Mark the pump, the workhouse and the brewery.

· for drawings and notes ·

GO WHERE YOUR THEORY SHOULD BREAK

1 · THE BREWERY



THEY DRANK BEER, NOT PUMP WATER

2 · THE WORKHOUSE



IT HAD ITS OWN WELL

3 · THE WIDOW MILES AWAY



SHE HAD PUMP WATER DELIVERED

AN ANOMALY IS YOUR THEORY'S EXAM

CANVAS · STAGE 03

Breaks it or supports it? Take each anomaly apart.

· for drawings and notes ·

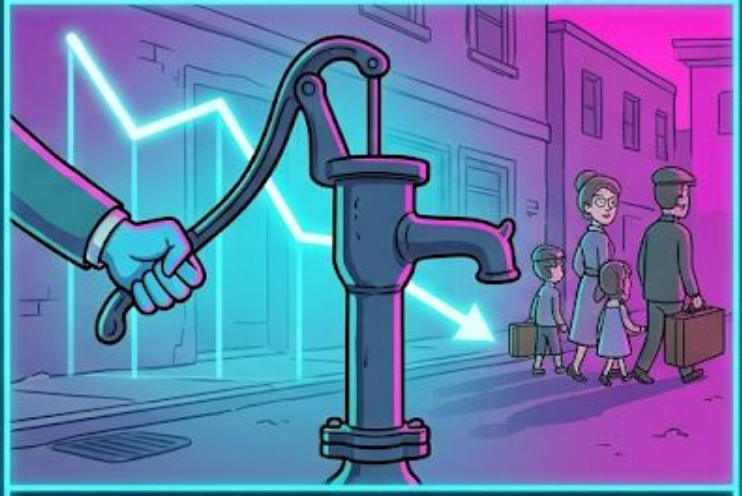
A WHOLE ERA HAD THE WRONG MAP

WHAT EVERYONE BELIEVED



DISEASE COMES FROM BAD SMELLS

THE GOOD-STORY TRAP



IT WAS ALREADY FADING

A GOOD STORY ISN'T PROOF

YOUR SUPERPOWER

WHERE? **WHO?** **WHEN?**

NO MICROSCOPE NEEDED



DON'T ASK WHAT YOUR MAP SAYS. ASK HOW IT COULD FOOL YOU.

CHOOSE THE NEXT CASE

CASE A · SPACE GERMS



DO GERMS GET MEANER IN ORBIT?

CASE B · THE RACE UP THE NERVE



A VACCINE THAT WORKS AFTER A BITE

CASE C · THE GENE THAT WON'T DIE



WHY A HARMFUL GENE SURVIVES

What backs this up

- **Primary source.** Snow J. *On the Mode of Communication of Cholera*, 2nd ed. London: John Churchill, 1855 — the map, the brewery, the workhouse (535 inmates, 5 deaths), the Hampstead widow, and Snow's own note that attacks had fallen before the handle came off.
- **Method and map** (door-to-door survey, one bar per death): Gavi; yourgenome.org.
- **The widow, Susannah Eley**: choleraandthethames.co.uk.
- **malaria = mala aria, "bad air"**: Etymonline; Wikipedia, "Miasma theory".
- **Broad Street outbreak, 1854** (~616 deaths; 500+ in ten days): Wikipedia, "1854 Broad Street cholera outbreak"; London Museum.
- **Brewery and workhouse**: Sick City Project; Wikipedia.
- **Well about 3 ft (0.9 m) from a leaking cesspit**: Wikipedia, "1854 Broad Street cholera outbreak".
- **Map image**: "Snow-cholera-map-1.jpg", Wikimedia Commons, public domain.

The microbe itself: Filippo Pacini described the cholera vibrio in Florence in 1854, and was ignored; Robert Koch identified it in 1883–84 and got the credit.